HG-DAgger with VR teleoperation

Recovering a household robot’s failures in the 2026 BEHAVIOR Challenge

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Recovery from the robot’s own failures

100 household tasks
20,000 expert demonstrations

R1Pro mobile robot in OmniGibson

Head and wrist cameras

π0.5-based policies

1. Run the policy

2. Save failures and simulator states

3. Restore a failure for an expert

4. Record a correction

One button switches full control



Quest 3S / Pico 4 Enterprise · CloudXR
Red border = human control

Play VR recovery (13 seconds)

Watch the policy

Press B to take over

Move the hands to correct

Press B to hand back

30 Hz control · 120 Hz physics

A record of who controlled every frame

intervention_mask
1 = human; 0 = policy

LeRobot v2.1 episodes

Only human intervals become human correction examples

Human action labels

Reached arm and torso positions

Commanded base velocity

Latched gripper command

Intervals shorter than 32 frames dropped

The recording format connects the VR interface to policy training.

Replay timing changed what people recorded

29 / 33

place recordings began with the can already held

Before: 5 s before the failure mark

People often recorded prevention.

After: 3 s past the mark

Changed after 78 of 82 recordings.

78 recordings in four days, on one task: human time limited collection.

Scripts and a stronger policy added corrections

Privileged scripts

4,166 recordings on 48 new tasks

AI and human review for selected skills
1,028 reviewed; 855 published

Stronger policy: pt50

754 successes in 3,961 attempts (19%)

500 first-round corrections kept
Run recipe lists 499

These experts take over saved failed states with an automatic gate.

A pick-up gain, with limits at task level

37.6% → 44.6%

Stronger-policy corrections

383 held-out pick-up phases

+7.0 percentage points

95% CI: +1.0 to +12.7

Matched demonstrations-only control;
10% corrections, 1,000 training steps

Human VR policy pilot: not trained

66 new-task human recoveries used for the phase-end detector (45 in training)

0 completed new tasks in 490 submitted-instance rollouts

Phase success does not establish an end-to-end task-score gain.

Interface decisions become dataset decisions

Clear control ownership and an explicit hand-back

A replay point chosen for recovery or prevention

Per-joint label checks before training

A user study of takeover design and latency

Source: team report, 3 Nov 2026, §§4–7. AI use: drafting and slide construction.